SHREE GURU TRADERS

Integrated Business Management & Accounting Software

Software Requirements Specification (SRS)
Developer-Ready Functional Requirements

Document field	Details
Business	Shree Guru Traders (SGT)
Document purpose	Requirements for designing, developing, testing, deploying and maintaining an integrated business software system.
Version	1.0 – Consolidated requirements
Date	30 October 2026
Status	Requirements baseline; unresolved business rules require approval before production configuration.

Core principle: One integrated transaction/accounting engine. Data entered once must update all applicable registers, ledgers, stock records, bills and reports without duplicate posting.

Document Control and How to Use This Specification

This document organizes the business requirements supplied for SGT into a structured specification for developers. It includes functional modules, business rules, data fields, workflows, calculations, security, reports, integrations, acceptance tests, and items that require owner/accountant confirmation.

The requirements below distinguish: (1) stated business requirements, which should be implemented; (2) recommended safeguards and standard software behavior; and (3) unresolved business/accounting choices that must not be guessed by the developer.

• Use this document for estimation, design, database planning, implementation, testing, and sign-off.

• Existing paper bills/registers and the supplied Cash/Credit Bill images should be used as visual references. Blurry or ambiguous labels must be confirmed from a clearer image or the business owner.

• The application must support daily operations and date-wise reporting, with a central company Khata Book that receives all applicable finalized transactions.

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1. Project Overview and Objectives

SGT operates agricultural commodity trading, commission (Dalali), purchases and sales, farmer settlements, and money-lending activities. The proposed system should replace disconnected paper registers and repetitive manual calculations with one integrated application.

1.1 Main objectives

• Record every transaction day by day with date, unique ID, party, amount, payment mode, and reference.

• Maintain a central company Khata Book and individual farmer, buyer, supplier, and other-party ledgers.

• Manage farmer registration, loans, interest, recoveries, commodity purchases, settlements, and outstanding balances.

• Manage Goods In, Goods Out, stock quantities, bags, weights, purchase values, and sales values.

• Generate Farmer Patti, Kharidi Patti, Dalali Vikri Patti, Dalali Kharidi Patti, and Cash/Credit Bills.

• Automatically update applicable ledgers, stock, cash/bank records, collection book, sales/purchase registers, and GST registers from finalized transactions.

• Provide reliable daily closing, audit history, error alerts, searchable history, and printable/exportable reports.

• Support secure use on desktop and mobile, with optional SMS notifications.

2. Scope and Users

2.1 In-scope modules

Module	Purpose
Dashboard and daily control	Live totals, alerts, day-end summary, pending work.
Farmer/party master	Farmer IDs, contact/profile, nominee, bank details, identity documents, signatures.
Loans and interest	Loan disbursement, calculation, repayment, settlement deduction, outstanding balances.
Commodity trading	Purchases, sales, commodity master, quantities, rates, deductions and stock.
Goods In / Goods Out	Date-wise stock movement and stock valuation.
Farmer Patti / Kharidi Patti	Farmer settlement and buyer-side commodity records.
Dalali	Vikri Patti and Kharidi Patti for commission transactions.
Kharidi	Kharidi Patti and Cash/Credit Bill workflows.
Khata Book/accounting	Central company ledger and party ledgers; integrated posting.
Cash, bank and cheque	Cash book, multiple bank accounts, transfers, cheques and reconciliation.
Registers	Sales, purchases, collections, GST, current accounts.
Expenses and commitments	Expenses, forms/fees, chit funds, insurance, investments and other obligations.
Calculator	General, interest, commission, hamali, settlement, weight and percentage calculations.
SMS and notifications	Optional event-based messages, delivery logs and reminders.
Audit and validation	Duplicate detection, reconciliation, controlled correction and error dashboard.

2.2 User roles (recommended)

Role	Typical permissions
Owner / Administrator	Full access, configuration, approvals, reports, user management, audit history.
Accountant	Ledgers, accounting, GST-related reports, bank reconciliation, approved corrections.
Trading / Purchase-Sales staff	Create commodity transactions and bills; limited access to sensitive farmer data.
Cashier	Cash receipts/payments, collections, cash closing; no access to unnecessary identity documents.
Read-only / Auditor	View permitted reports and audit trails without editing.

Permissions should be configurable by role and, where necessary, by individual user. Sensitive farmer identity, signature, and bank information must not be visible to every employee by default.

3. Core Architecture and General Rules

• Single source of truth: modules should use shared master data and a central transaction/accounting engine.

• Enter a business event once. The system should generate all related register, ledger, stock, and report updates automatically.

• Every record must have an immutable internal ID and a human-readable reference number where relevant.

• Number series must be configurable by document type and financial year. Example formats: SGT-LN-000001, SGT-FP-000001, SGT-KP-000001, SGT-PY-000001.

• Do not reuse issued numbers. Cancelled records remain visible with cancellation status and reason.

• Use the actual transaction date and separately record creation time, last update time, and user.

• Draft transactions do not affect final accounting or stock until posted/finalized, unless a workflow explicitly defines a reservation or pending status.

• Finalized transactions must not be silently overwritten or deleted. Corrections should use controlled edit/reversal/adjustment workflows and audit logs.

• Use decimal-safe financial calculations, configurable rounding, and consistent units. Store original quantity/rate and calculated values.

• Support attachments such as bills, receipts, bank proofs, identity documents, and signatures with access restrictions.

• Provide search by name, ID, document number, date, commodity, payment reference, and status.

• A failed SMS, printer, or export operation must not undo a successfully saved business transaction.

4. Master Data and Farmer Registration

4.1 Farmer profile

Field	Requirement
Internal Farmer ID	Auto-generated, unique, never reused.
Farmer name	Required; searchable.
Village/address	Village and full address fields as required.
Contact details	Phone number(s); consent/preferences for SMS where applicable.
Nominee	Name, relationship, contact and other necessary details.
Bank details	Bank name, branch, account holder, account number and IFSC where needed; masked in normal screens.
Identity proof	Document type, number if needed, uploaded file/scan and verification status.
Farmer ID	Government-issued or other identifier, if applicable; type must be specified.
Signature	Stored signature image/document or signed form, with restricted access.
Status	Active/inactive; reason and date for deactivation.
Audit data	Created/updated by, timestamps, attachments, verification notes.

Before a loan is issued, the farmer must have a registered profile and pass required-field validation. The system should warn about possible duplicates based on name, phone, identifier, village, and bank details without automatically merging profiles.

4.2 Other party masters

• Maintain master profiles for buyers, wholesalers, factories, traders, suppliers, transporters, employees, service providers, government departments, banks, insurers, chit-fund providers, and other parties.

• Party profiles should include opening balance, contact/address, tax identifiers where applicable, payment terms, status, notes, and linked transaction history.

• Company profile settings should hold SGT legal name, address, contact numbers, GSTIN, PAN and other bill details centrally so that changes are applied consistently.

5. Loan and Interest Management

5.1 Loan disbursement

A loan can be issued only to a registered farmer unless an authorized exception is configured. Support cash and bank transfer.

Field	Requirement
Loan ID	Auto-generated unique reference.
Farmer	Linked Farmer ID and profile.
Disbursement date	Actual date funds are paid.
Principal amount	Required positive amount.
Payment mode	Cash, bank transfer, cheque or another configured mode.
Cash/bank account	Required according to payment mode.
Bank reference	UTR/transaction reference or cheque details when applicable.
Terms	Interest rate, method, frequency, start/end rules and repayment terms, once approved.
Remarks/attachment	Purpose, approval, receipt or supporting document.

• On finalization, create the loan record, cash/bank movement, farmer ledger entry, company Khata Book posting and printable receipt/voucher.

• Support multiple loans per farmer and a consolidated farmer loan summary.

• Record repayments separately and allocate amounts between interest and principal according to the approved policy.

5.2 Interest calculation

• Interest must start from the actual disbursement date, using the approved interest method and configured rate.

• Show principal outstanding, interest accrued, interest received, principal repaid, total due, last calculation date, and transaction history separately.

• Allow daily/monthly/yearly rate units and start/end dates in the calculator, but do not post calculator-only results to accounts unless the user confirms an accounting transaction.

• When a farmer settlement is prepared, calculate interest up to the approved settlement date and show the calculation detail.

• Changing a rate or method must be permission-controlled, dated, and audited. Historical calculations must be reproducible.

Important: the business owner/accountant must approve the interest rate, simple/compound method, day-count convention, treatment of partial repayments, grace periods, rounding, and whether unpaid interest itself earns interest. Developers must not guess these rules.

6. Commodity Trading and Stock

6.1 Commodity master

Initial commodities include sunflower, maize, cotton, paddy, soybean, groundnut/groundnuts, alsandi, ragi, kharadi, safflower and kadli. The list must be editable so authorized users can add or deactivate commodities.

• Each commodity should have a standard name/code, allowed units, optional grade/variety, HSN where applicable, and configurable stock-tracking rules.

• Support bags, kilograms, quintals and tonnes, with explicit conversion settings. Do not assume every bag has the same weight.

6.2 Purchase and sale transactions

• Record date, document number, supplier/farmer or buyer, commodity, bags, gross/tare/net weight where used, quantity, unit, rate, value, deductions, charges, tax treatment, payment status and payment reference.

• Support cash, credit, partial payments and pending amounts.

• Support purchases from farmers and other traders, and sales to wholesalers, traders, factories, farmers and individual customers.

• Maintain transaction-level traceability to bills, pattis, parties, stock movements and accounting postings.

• Allow configured quality/weight adjustments such as moisture, impurities, weighing difference, hamali, transport and other charges where applicable.

• Warn if a sale would exceed available stock. Authorized adjustments require a reason and audit trail.

6.3 Goods In / Goods Out stock register

Movement	Minimum fields	Automatic effect
Goods In	Stock Entry ID, date, commodity, supplier/farmer, bags, weight (Q/kg), rate, total purchase value, charges/deductions, payment status, related bill, notes.	Increase stock on finalized receipt; update purchase/stock records and applicable ledger/accounting entries.
Goods Out	Goods Out ID, date, commodity, buyer, bags, weight, rate, sales value, dispatch/vehicle details, payment status, related bill, notes.	Decrease stock on finalized dispatch/sale; update sales/stock records and applicable ledger/accounting entries.
Stock adjustment	ID, date, commodity, quantity/weight change, reason, approver, attachment.	Adjust stock only with permission and an auditable reason.

Formula: Closing stock quantity = Opening stock + Goods In – Goods Out ± approved adjustments. Track bags and weight separately. Stock value must use an owner/accountant-approved valuation method.

Dalali transactions performed on behalf of a farmer must be marked as commission/agency transactions where appropriate and must not automatically be treated as SGT-owned stock. The system must distinguish owned stock, consignment/agency goods and goods merely handled for a customer.

6.4 Stock reports

• Date-wise goods-in/goods-out, commodity-wise stock, opening/closing stock, bags and weight, purchase value, sales value, low-stock alerts, stock adjustment history, and negative-stock exceptions.

• Support search by date, commodity, supplier, buyer, bill number and movement ID.

7. Bills, Patti and Farmer Settlement

7.1 Farmer Patti

Farmer Patti is the farmer settlement statement for commodity delivered by a farmer and sold/settled through SGT.

Field / calculation	Requirement
Identification	Auto Patti number, date, Farmer ID/name, crop/commodity, related sale/buyer references.
Quantity	Total bags, quintals, and any additional weight unit required.
Crop value	Total eligible quintals × farmer settlement rate per quintal.
Commission	Crop value × configured commission percentage ÷ 100.
Hamali	Number of bags × configured fixed hamali charge per bag.
Other deductions	Hand payment/advance already given, vehicle rent, tukad fee, delivery or other approved deductions.
Total deductions	Sum of applicable commission, hamali and other deductions, with each item separately visible.
Amount before loan adjustment	Crop value – total deductions.
Loan recovery	Outstanding principal and accrued interest shown separately and deducted according to approved settlement rules.
Net farmer payable	Amount before loan adjustment – approved loan recovery, with insufficient-settlement handling.

Field / calculation	Requirement
Signatures/print	Farmer/authorized signatures as required; printable/PDF bill.

• Rates for commission and hamali must be configurable by commodity, transaction type, party or date if the owner requires this.

• If the amount available is less than the farmer's outstanding loan and interest, do not hide the shortfall or create an unexplained negative payment. Show net payable as zero where appropriate and carry forward the remaining liability according to approved accounting rules.

• When the Patti is finalized, create linked farmer/buyer records, stock movement where SGT owns the goods, loan recovery entries, cash/bank payment if paid, and company Khata Book postings.

• Editing or reprinting a finalized Patti must not deduct the loan twice or create another stock movement. A revised/voided Patti must reverse or adjust linked postings transparently.

7.2 Buyer-side Kharidi Patti

• Capture auto bill number, date, buyer, commodity, bags, quintals/weight, selling rate, total amount, deductions/tax treatment where applicable, payment terms and signature.

• Support buyer-wise bill history, partial receipts, outstanding amount and statement.

• Link buyer-side transaction to the corresponding farmer settlement or stock purchase where applicable, but allow farmer rate and buyer selling rate to differ.

The business owner must confirm whether a given buyer-side Kharidi Patti is a purchase, sale, or agency settlement document in each workflow. The document label alone must not determine the accounting treatment.

8. Dalali Section

SGT has a Dalali/commission section. The system must support the two bill formats below and preserve the layout of the existing paper bills as closely as practical.

8.1 Dalali Vikri Patti

• Fields: firm name/address, auto bill and serial number, date, farmer name, buyer name, class/serial number if used, bags/andiges, crop, weight (quintals and grams/other required units), rate, total crop amount.

• Expense lines: commission, hamali, tukad fee, delivery, “niitu fee” (label/spelling and meaning to be confirmed), other charges, total expenses, amount/deductions, final total and customer signature.

• Calculations, order of deductions, and print layout should follow the existing paper format after the owner confirms the exact meaning of each field.

• Auto-number, save draft/final, search, reprint, export PDF, and link to party ledger and Khata Book.

8.2 Dalali Kharidi Patti

• Fields: firm name/address, bill/serial number, date, buyer, crop, bags/andiges, total weight (kg and grams or configured units), rate, gross value.

• Deduction/tax lines: tukad fee, RMC, commission, CGST, SGST, other approved charges, total amount and merchant signature.

• Tax and deductions must be configured based on the actual transaction and accountant-approved rules; the system must not apply GST automatically to every commodity or transaction.

• Support printable/PDF output, party ledger posting, buyer outstanding tracking and link to stock movement only where appropriate.

9. Kharidi Section and Cash/Credit Bill

9.1 Kharidi Patti

• Fields: company name/address, farmer name/address, crop, quantity in quintals and kilograms, unit price, total value, date, auto bill number, SGT/merchant signature and farmer signature.

• Generate printable/PDF bill and link to purchase register, stock Goods In, farmer/supplier ledger, payment records and company Khata Book.

9.2 Cash/Credit Bill

Use the user-provided existing Cash/Credit Bill image(s) as a layout reference. The developer must inspect the clearest original and confirm blurry or ambiguous labels before finalizing the print template.

• Company details: firm name, address, GSTIN, PAN, contact numbers and any other approved legal details.

• Invoice number/date; customer name/address; customer GST number where applicable; lorry/vehicle number; goods/commodity description; rate; quantity/bags/weight; amount; delivery number; origin/destination; total; amount in words.

• Transport details where applicable: lorry owner, driver name, driving licence number, vehicle number; driver and authorized signatures.

• Include the approved terms and conditions/payment/interest text from the existing form, once confirmed.

• Support cash and credit invoices, partial/full receipts, outstanding balances, cancellations/credit notes where approved, search/reprint, PDF and print.

• The field read as “CST” in the supplied image may be ambiguous or legacy terminology. Confirm whether it should remain or be replaced by an appropriate current tax/charge field. Do not assume.

• Finalized invoices must update the appropriate sales/purchase register, party account, collection register, stock if applicable, GST register if applicable, and central Khata Book.

10. Khata Book and Central Accounting Engine

10.1 Required ledgers

• Company Khata Book: consolidated business transaction history and accounting entries.

• Farmer-wise Khata: loans, interest, principal repayments, commodity purchases/settlements, advances, payments and other transactions.

• Buyer-wise Khata: bills, receipts, partial payments, adjustments, refunds and outstanding balances.

• Supplier/trader/transport provider and other party ledgers.

• Cash book and separate bank ledgers.

• Expense/income, owner capital/withdrawal, tax and other required control accounts.

10.2 Automatic posting

Every applicable finalized transaction in every module must post automatically to the relevant ledgers. Users should not manually re-enter the same event into the company Khata Book.

Business event	Expected linked updates (as applicable)
Loan disbursement	Loan register, farmer ledger, cash/bank, company Khata Book, receipt.
Loan repayment	Farmer loan/interest allocation, cash/bank, farmer ledger, company Khata Book.
Commodity purchase	Purchase register, Goods In, supplier/farmer ledger, payment status, accounting ledger.
Commodity sale	Sales register, Goods Out where SGT stock is sold, buyer ledger, collection/outstanding, accounting ledger.
Farmer Patti	Settlement, farmer payable, commission/charges, loan recovery, payment, stock/agency records as applicable.
Buyer collection	Collection book, buyer balance, cash/bank, invoice settlement, company Khata Book.
Expense	Expense register, payee ledger if used, cash/bank, company Khata Book.
Bank transfer	Decrease one cash/bank account and increase the other; not income or expense.
GST-related invoice	Tax register/control accounts where applicable, subject to approved tax rules.

10.3 Accounting and balance safeguards

• Use an accounting design that supports balanced double-entry postings where appropriate; do not rely on a single ambiguous “credit/debit” field for every financial concept.

• Show both business/accounting debit-credit and cash-in/cash-out labels clearly where needed.

• Keep principal, interest accrued, interest received, farmer payable, buyer receivable, income, expense, stock and tax balances distinct.

• Do not classify internal cash-to-bank transfers as income/expense.

• Do not count loan principal as interest income; do not count loan disbursement as an operating expense without an approved accounting policy.

• Record the source document ID on every posting and enforce idempotency so one source event cannot post twice.

• Opening balances must be entered and approved during setup; include a documented opening-balance date and source.

11. Daily Operations and Day-End Closing

11.1 Day-by-day transaction capture

All business activity must be recorded with the transaction date and unique ID. The software must provide a date-wise register and allow authorized backdated entries with clear audit metadata.

• Opening cash, bank, party, stock, loan and other balances should be captured during setup as appropriate.

• Daily transactions include purchases, sales, Goods In/Out, loans, interest events, collections, payments, Patti, expenses, bank movements, forms/fees, chit fund, insurance and other recorded commitments.

• The software should update the central Khata Book and applicable ledgers/registers from each finalized transaction.

11.2 Daily Credit/Debit and cash view

Provide a daily transaction view showing date, transaction ID, description, party, source document, accounting debit/credit where relevant, cash/bank money-in or money-out, and resulting balance. These are different concepts and should be labeled separately.

Cash balance formula: Opening cash + cash receipts – cash payments ± approved adjustments. Bank book balance is calculated separately for each bank account.

11.3 Day-end report and close

• Show opening and closing cash/bank balances, receipts, payments, purchases, sales, stock movements, loans issued, repayments, interest accrued/received, commission, expenses, buyer receivables, farmer/supplier payables, stock balances and unresolved errors.

• Allow a daily closing checklist and optional owner approval. A day-end report should be printable/exportable to PDF/Excel.

• After day close, flag changes to that date. Authorized reopening/correction must record who, when, why, and what changed.

• Recalculate affected balances consistently after an approved correction, preserving previous values in audit history.

11.4 Date-wise history

• Search any single date or date range across company Khata Book, individual ledgers, cash/bank, bills, collection book, purchases/sales, stock and expenses.

• Allow drill-down from daily totals to source bills and individual transactions.

12. Cash, Bank and Cheque Management

12.1 Bank accounts

• Support multiple business bank accounts, each with bank name, branch, account holder, masked account number, IFSC, account type, opening balance, current book balance and active/inactive status.

• Do not store bank passwords, PINs, OTPs, internet banking credentials or card security codes.

12.2 Bank transaction register

• Fields: transaction ID, date, bank account, credit/debit, description, party, UTR/reference, cheque number if applicable, amount, bank charges, running book balance and remarks.

• Book balance = opening balance + credits – debits. Show book balance separately from actual bank statement/available balance.

• Automatically link loans, buyer receipts, farmer settlements, supplier payments, expenses, bank charges and other events.

12.3 Cheque management

• Track issued/received cheque number, date, bank, party, amount, expected clearing date, status (issued, received, pending, cleared, bounced, cancelled) and linked transaction.

• Do not treat a cheque as cleared cash until the configured clearing event is confirmed. Bounced/cancelled cheques require reversal/adjustment with a full audit trail.

12.4 Reconciliation

• Allow bank statement import where a supported format is available; match by amount/date/reference and flag unmatched, duplicate, missing or differing entries.

• Track pending cheques, bank charges, bank interest and timing differences.

• Provide daily/monthly/yearly statements, credit/debit totals, cheque reports, bank charges and cash-vs-bank reports.

13. General Transactions, Expenses and Commitments

13.1 General money transaction

For transactions not handled by a dedicated module, provide a controlled general-entry form with transaction ID, date, party/organization, category, money in/out/transfer, amount, mode (cash/bank/cheque), account, reference, purpose, related bill/loan, remarks and attachment.

13.2 Business and owner account

• Track capital introduced, owner withdrawals, commission income, interest income, commodity trading activity, operating expenses, bank charges, transfers and other approved accounts.

• Separate personal expenses from business expenses where required by the owner/accountant.

• Transfers between SGT's own cash/bank accounts must be recorded as transfers, not income or expenses.

13.3 Expenses

• Configurable categories include wages/salaries, shop rent, utilities, hamali, transport/fuel/delivery, office expenses, banking charges, repairs, government fees, insurance, chit fund-related payments, loan interest paid and others.

• Each expense includes expense ID, date, category, payee, amount, payment mode/account, receipt, tax details where applicable and remarks.

13.4 Forms and government fees

• Record application/form name, department, applicant/beneficiary, reference number, submission date, fee, payment date/mode, receipt, status, attachments and due-date reminders.

13.5 Chit funds

• Record company/group/reference, total value, monthly instalment, number of instalments, due dates, paid/pending amounts, total contributed, amount received, auction/dividend if applicable, payment mode, receipt, documents and status.

• Classification as expense, investment, receivable or other account must be configurable/approved; do not automatically treat every contribution as an expense.

13.6 Insurance

• Record insurer, policy number/ID, type, policyholder, premium, frequency, start/renewal dates, sum insured, paid/pending amount, mode, status, documents and reminders.

• Restrict access to personal insurance information.

13.7 Other assets and commitments

• Allow flexible records for fixed/recurring deposits, investments, personal or business loans given/received, vehicle/property payments, advances/refundable deposits, memberships and other commitments.

• Dashboard must distinguish assets, liabilities, income, expenses, cash, receivables and commitments.

14. Sales/Purchase, Collection and GST Registers

14.1 Sales and purchase register

• Record invoice number/date, party, GSTIN where applicable, transaction type, commodity/description, quantity, taxable amount, applicable tax and total.

• Auto-populate from finalized bills; avoid duplicate manual entries. Provide daily/monthly/yearly summaries and PDF/Excel export.

14.2 Collection book

• Track buyer/customer, address, GSTIN if applicable, invoice number/date, amount due, payment received, date/mode, reference and remaining balance.

• Support partial and full payments, multiple receipts against one bill, buyer statements, overdue lists and collection summaries.

14.3 GST register

• Fields as applicable: invoice date/number, party/GSTIN, commodity, HSN, taxable value, CGST, SGST, IGST, other tax/charge, total tax and invoice total.

• Provide period-wise summaries, reconciliation and exports for the accountant.

• GST applicability, HSN, rates, exemptions and invoice treatment must be reviewed and configured by the business's qualified accountant/tax adviser. The software is a recordkeeping tool and must not promise legal/tax compliance by itself.

• The existing bill image includes a possibly ambiguous “CST” label; confirm the intended field before reproducing it.

15. Calculator and Automatic Calculations

15.1 Built-in calculator

• Available from all main screens without closing or losing the current form.

• Basic operations: addition, subtraction, multiplication, division, percentage, decimals, clear/reset and calculation history.

15.2 Interest calculator

• Inputs: principal, rate, rate unit, start date, end date, calculation method, repayment details if applicable.

• Outputs: duration, interest, total amount and calculation breakdown.

• Calculator results remain non-posting until the user explicitly creates/approves a transaction.

15.3 Other automatic calculations

• Commodity value = quantity × unit rate, using a clearly defined unit.

• Commission = configured eligible value × commission percentage ÷ 100.

• Hamali = eligible number of bags × charge per bag.

• Weight conversion based on configured standard units; show source and target units.

• Farmer Patti deductions, net payable, loan recovery, buyer outstanding, expense totals, profit reports and percentage calculations.

• Settings for rates, rounding and units must be permission-controlled and changes audited.

16. Live Dashboard, Alerts and SMS

16.1 Dashboard

• Live totals for opening/cash/bank balances, loans outstanding, accrued interest, farmer dues, buyer dues, purchases/sales, collections/payments, commission, expenses, stock and unresolved errors.

• Clicking a total should open its underlying transaction list where the user has permission.

16.2 Alerts

• Configurable alerts for large loans/payments, overdue loans or buyer dues, failed SMS, unusual adjustments, suspected duplicates, cash mismatches, stock shortage and approval-required changes.

• Alert thresholds and recipients must be configurable.

16.3 SMS integration

SMS requires a compatible third-party provider, account setup, consent/notification policy, and cost approval. SMS is an additional communication record, not a replacement for an official bill, receipt or bank confirmation.

Event	Possible message
Farmer registration	Farmer ID created/registered confirmation.
Loan disbursement	Loan amount, date, reference and contact information.
Interest update	Interest calculation summary as of a stated date, if enabled.
Commodity received / Farmer Patti	Quantity, bill reference, settlement summary and payment status.
Loan recovery / farmer payment	Amount applied to principal/interest and remaining balance, if approved.
Buyer bill / Kharidi Patti	Invoice reference, amount and due date.
Buyer payment	Amount received, reference and remaining outstanding.
Overdue reminder	Outstanding amount and contact details for follow-up.

• Store message ID, recipient, linked transaction, template/content, sent time, provider, delivery status and failure reason.

• If SMS fails, the transaction remains saved. Retrying must not duplicate accounting entries or unintentionally send duplicate messages.

• Templates should be editable and support Kannada/English Unicode if the provider supports it.

• Do not include full bank account numbers, identity numbers or unnecessary sensitive information in SMS.

17. Error Detection, Audit and Corrections

17.1 Automatic validation

• Missing required fields, dates, rates, quantity, party, bill number, required documents or signatures.

• Possible duplicate farmers, loans, bills, payments, collection entries and invoice numbers.

• Incorrect commodity total, commission, hamali, rent, deductions, settlement, interest, loan balance, GST or running balance.

• Missing links between Farmer Patti, Kharidi Patti, cash/credit bill, collection, GST, stock and cash/bank records.

• Negative or insufficient stock, sales exceeding available stock, incorrect unit conversion, or missing stock movement.

• Unmatched linked records, duplicate posting, incorrect loan-recovery allocation, double deduction or unexplained remaining debt.

• Cash/bank reconciliation mismatch and accounting entries that do not balance where double-entry applies.

17.2 Error dashboard

• Each issue should show severity (critical/high/medium/low), description, date, linked party/bill/transaction, suggested action, assigned user, status (open/review/resolved) and resolution notes.

• Provide daily error report as PDF/spreadsheet.

17.3 Audit trail

• Record original and corrected values, user, timestamp, reason, approval, and all linked balance recalculations.

• Finalized records must not be silently deleted. Use cancellation, reversal or adjustment entries with references to the original.

• High-value corrections, loan-rate changes, stock adjustments and day reopening should support approval rules.

• Maintain system logs and avoid exposing sensitive values in logs.

18. Security, Backup and Non-Functional Requirements

18.1 Security and privacy

• Secure login, password hashing, session expiry and optional multi-factor authentication.

• Role-based access and least-privilege permissions.

• Encryption in transit and appropriate encryption at rest, especially for identity documents, signatures, bank details and backups.

• Mask bank account and identity numbers on ordinary screens; restrict viewing/downloading of sensitive documents.

• Record login, important data access, exports, configuration changes and financial corrections where practical.

• Never store bank passwords, PINs, OTPs or internet-banking credentials.

• Define retention, consent, access, deletion and breach-handling processes consistent with applicable law and business policy.

18.2 Reliability and usability

• Desktop-first responsive interface that also works on supported mobile devices; confirm whether offline operation is required.

• Fast search, date filters, clear validation messages, readable bill printouts and a consistent date/currency format.

• Use India-specific formatting where appropriate: ■, Indian number grouping, day-month-year display, and Asia/Kolkata business timezone.

• Prevent concurrent edits from silently overwriting one another; use record locking/version checks for sensitive finalized transactions.

• Provide monitored backups, restore testing, error logging, recovery procedures and documented maintenance.

• Do not claim a transaction is saved until the server/database confirms successful persistence.

18.3 Backup and disaster recovery (recommended baseline)

• Automated encrypted daily backup plus more frequent database backups according to transaction volume.

• Maintain a separate protected backup location and define retention period with the owner.

• Test restoration periodically and document acceptable recovery time and maximum data-loss window.

• Owner must approve hosting, data location, retention, recovery targets and responsibility for backup monitoring.

19. Reports and Exports

Report	Minimum content
Daily business summary	Opening/closing balances, receipts, payments, purchases/sales, expenses, loans, stock and alerts.
Company Khata Book	Date, transaction/reference, narration, debit/credit, account and running balance.
Farmer statement	Opening balance, loans, interest, repayments, Patti, deductions, payments and closing balance.
Buyer statement	Opening balance, invoices, receipts, adjustments, outstanding and closing balance.
Loan/interest report	Loan-wise principal, interest accrued/received, repayment history and outstanding.
Goods In/Out and stock	Commodity, bags, weight, dates, movement, opening/closing quantity and value.
Sales/purchase register	Bill, date, party, commodity, quantity, value, applicable tax and total.
Collection/outstanding	Invoice-wise due, receipts, partial payments, overdue age and balance.
Cash/bank/cheque	Account, date, credits, debits, references, status, running balance and reconciliation.
Expense/commitment	Category, payee, date, amount, payment status, due dates and attachments.
GST support	Period-wise invoice and tax data, subject to accountant-approved rules.
Audit/error report	Corrections, approvals, duplicate warnings, mismatches and resolution status.

• Reports should support date ranges, party, commodity, document type, status, and export to PDF/Excel where relevant.

• Export permissions should be restricted because exports may contain sensitive data.

20. Recommended Workflows

20.1 Farmer loan workflow

• Search farmer; create and verify profile if new.

• Enter loan amount, date, payment mode, account/reference and approved interest terms.

• Validate mandatory fields and approval threshold.

• Finalize disbursement and generate loan ID and receipt.

• Automatically update loan register, farmer ledger, cash/bank and company Khata Book.

• Send SMS if enabled; log success/failure without affecting the saved transaction.

20.2 Purchase / Goods In workflow

• Select supplier/farmer and commodity; enter date, bags, weight, rate, deductions and payment terms.

• Review calculated value and unit conversions.

• Finalize purchase/receipt.

• Create one linked Goods In movement and update stock, purchase register, party account, payment status and accounting postings as applicable.

• Print bill/Patti and reconcile payment later if credit or partial payment.

20.3 Farmer settlement workflow

• Select farmer, commodity lot, quantity and approved settlement rate.

• Calculate crop value, commission, hamali and other deductions.

• Calculate outstanding principal and interest up to the approved settlement date.

• Display gross value, every deduction, loan recovery, remaining loan balance and net amount payable.

• Require user review/approval; finalize settlement.

• Post linked entries exactly once, update stock/agency records as applicable, generate printable Farmer Patti, record actual payment when made, and notify farmer if enabled.

20.4 Buyer bill and collection workflow

• Create/finalize bill with customer, commodity, quantity, rate, applicable tax and payment terms.

• Update buyer ledger, sales register, stock where applicable, GST register where applicable and company Khata Book.

• Record each partial/full receipt with date, mode and reference.

• Update collection book, cash/bank and outstanding balance; send receipt/notification if enabled.

20.5 Daily close workflow

• Review all transactions for the selected business date.

• Resolve or acknowledge validation errors and reconciliation differences.

• Compare physical cash and bank records with book balances; record authorized adjustments with reasons.

• Review stock movement, bills, collections, expenses and pending payments.

• Generate daily closing report and close the day, optionally with owner approval.

• Any later change requires an authorized reopen/correction and audit trail.

21. Data Model Overview (Developer Guidance)

The final database design is the developer's responsibility, but the following entities and relationships are required. Use primary keys, foreign keys, indexes, transaction-safe writes and unique constraints.

Entity / table concept	Key fields and relationships
User / Role / Permission	User ID, role, permissions, status, last login.
Company Settings	Legal details, document series, rates, units, tax configuration, fiscal year.
Party / Farmer	Party ID, Farmer ID, profile, contacts, documents, bank/nominee details.
Commodity / Unit	Commodity ID, name, unit conversions, grade, tax metadata if applicable.
Loan / Loan Transaction	Loan ID, farmer ID, disbursement, terms, repayment allocations, balances.
Bill / Bill Line	Bill ID, type, party, date, lines, quantities, rates, deductions, tax, status.
Stock Movement	Movement ID, commodity, quantity/bags/weight, direction, source document, status.
Ledger Account / Journal Entry / Journal Line	Account, source event, date, debit/credit, balanced posting, references.
Cash/Bank Transaction	Transaction ID, account, date, mode, amount, reference, status.
Collection / Payment	Receipt/payment ID, party, bill/loan link, allocation, mode, reference.
Expense / Commitment	Category, payee, date, amount, due dates, attachments, status.
GST / Sales / Purchase Register	Invoice references, party, commodity, taxable values and applicable taxes.
SMS Log	Provider ID, recipient, linked event, template, status, retry data.

Entity / table concept	Key fields and relationships
Audit Log / Error Issue	Entity, record ID, before/after, user, time, reason, status.
Attachment	File reference, type, access policy, owner record, checksum/metadata.

Financial and stock postings should be committed atomically with the source transaction or handled by a reliable retry/outbox mechanism. A repeated request or retry must not create duplicate ledger lines or stock movements.

22. Acceptance Criteria and Test Cases

The developer should demonstrate these scenarios in a staging/test environment using sample data before production sign-off.

ID	Test	Expected result
AT-01	Register a farmer with required details.	Unique Farmer ID created; profile searchable; sensitive data access restricted.
AT-02	Issue a cash loan.	Unique Loan ID and receipt; cash balance decreases; farmer loan/Khata entries appear once.
AT-03	Issue a bank-transfer loan.	Correct bank account and reference recorded; farmer and company ledgers updated.
AT-04	Calculate interest for a chosen period.	Result matches the approved rate/method/day-count policy and shows a breakdown.
AT-05	Finalize commodity purchase.	Purchase register and Goods In update once; stock quantity/weight increase correctly.
AT-06	Finalize commodity sale from owned stock.	Sales register and Goods Out update once; stock reduces; insufficient stock is flagged.
AT-07	Record Dalali transaction for farmer-owned goods.	Agency/commission treatment is used; SGT-owned stock is not incorrectly changed.
AT-08	Generate Farmer Patti with loan recovery.	Commission, hamali, other deductions, principal and interest are itemized; recovery is not duplicated.
AT-09	Record partial buyer payment.	Collection book and buyer balance update; remaining due is correct.
AT-10	Record bank transfer between SGT accounts.	Both account balances update; no income/expense is created.
AT-11	Finalize applicable GST invoice.	Approved tax fields and registers update correctly; non-applicable transactions are not taxed by default.
AT-12	Retry SMS after provider failure.	Saved transaction remains intact; no duplicate accounting posting; retry is logged.
AT-13	Attempt duplicate bill/transaction submission.	System prevents duplicate posting or flags duplicate according to configured rule.
AT-14	Correct a finalized transaction.	Authorized correction/reversal is recorded; original and change history remain visible.
AT-15	Close and later reopen a day.	Closing report is retained; reopening and changes require permission and are audited.
AT-16	Run date-wise company Khata Book.	All applicable finalized transactions for the chosen date appear with linked references.
AT-17	Reconcile bank statement.	Matched and unmatched items are identified; no automatic deletion of unmatched records.
AT-18	Restore a backup.	Database and attachments can be restored according to the agreed recovery procedure.

22.1 Sign-off conditions

- All critical acceptance tests pass.
- Opening balances and document numbering are validated by the owner/accountant.
- Paper bill layouts and required labels are approved.
- Interest, commission, hamali, units, stock valuation, GST, and accounting policies are approved.
- User permissions, backup restoration, export access and audit trails are demonstrated.
- A documented user guide, admin guide, deployment details and support/maintenance plan are delivered.

23. Implementation Phases

Phase	Recommended deliverables
Phase 1 – Discovery and approval	Confirm paper forms, account rules, interest/commission rates, units, tax handling, opening balances, roles and reports.

Phase	Recommended deliverables
Phase 2 – Core platform	Login/roles, company settings, master data, numbering, audit log, database and backup foundation.
Phase 3 – Ledgers and cash/bank	Party ledgers, company Khata Book, opening balances, cash book, bank accounts, payments/receipts and reconciliation.
Phase 4 – Farmers and loans	Farmer registration, documents, loans, interest engine, repayments and receipts.
Phase 5 – Trading, bills and stock	Commodity master, purchase/sales, Goods In/Out, Farmer Patti, Kharidi Patti, Dalali forms and Cash/Credit Bill.
Phase 6 – Registers and commitments	Collection, sales/purchase, GST support, expenses, forms, chit funds, insurance and other records.
Phase 7 – Dashboard, SMS and validation	Live dashboard, alerts, SMS provider, duplicate/error checks, daily closing.
Phase 8 – Testing and rollout	Migration/opening balances, user acceptance tests, training, backup restore test and production launch.

The developer should estimate each phase separately and identify dependencies, third-party fees, hosting costs, SMS costs, support charges and any items outside the agreed scope.

24. Decisions Requiring Confirmation

These points are intentionally left for the owner/accountant to confirm. The developer must not invent business rules for them.

Topic	Decision needed
Interest policy	Rate(s), simple/compound, daily/monthly/yearly basis, day-count convention, repayment allocation, unpaid interest treatment, rounding.
Commission and hamali	Rates by commodity/transaction/party, effective dates, calculation base and whether deductions are taxable.
Quantity and weights	Bag conventions, quintal/kg/gram conversions, gross/tare/net weight, rounding and allowable tolerances.
Stock ownership	Which transactions are SGT-owned stock vs farmer-owned/agency/Dalali goods; stock valuation method.
Farmer Patti	Exact deduction order, treatment of hand payments/advances, vehicle rent, loan shortfall, residual debt and net payment rules.
Dalali labels	Exact meaning/spelling of “niitu fee”, “tukad fee”, “RMC”, “class/serial number”, “andiges”, and exact calculation order.
Bill layout	Clear copies of all paper forms; confirm ambiguous Cash/Credit Bill field such as “CST”, delivery/origin/destination and terms.
GST/accounting	Accountant-approved applicability, HSN, rates, tax fields, credit/debit notes, journal treatment and statutory report needs.
Financial year	Financial year start/end, numbering reset rules and backdated-entry policy.
Opening balances	As-of date and amounts for cash, each bank, farmer/buyer/supplier accounts, loans, stock and other accounts.
Payment handling	Whether payments can be split across cash/bank/cheque, allocation rules, cheque clearing and bounced cheque process.
SMS	Provider, sender ID/template approval, languages, consent, event selection, message cost budget and retry rules.
Hosting and offline	Cloud/local hosting, permitted devices, internet reliability, offline requirement and sync behavior.
Access and approvals	User list, roles, approval thresholds for loans, rate changes, corrections, stock adjustments and day reopening.
Data retention	Backup frequency, retention period, data location, recovery targets and document retention policy.
Reports and language	Required report formats, Kannada/English interface and printed bill language.

Final Developer Instruction

Build SGT as one integrated, auditable business system—not as separate spreadsheets or disconnected modules. Every finalized business event must update all applicable records exactly once, with clear source references, correct date-wise history, reliable balances, stock movement where appropriate, and controlled corrections. Any uncertain accounting, tax, interest, or paper-form rule must be confirmed with the owner/accountant before it is hard-coded.